

HAMHUNG, Korea, Democratic People's Republic of

WMO: 470410

Lat: 39.933N

Lon: 127.550E

Elev: 22

StdP: 101.06

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.8	-11.0	-24.4	0.4	-8.6	-21.9	0.5	-7.5	9.5	-7.1	8.9	-7.4	1.9	50	0.387

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.9	32.2	24.4	30.4	23.8	28.9	22.9	26.2	30.0	25.2	28.7	24.4	27.4	3.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.1	20.3	28.5	24.1	19.1	27.3	23.4	18.3	26.4	81.8	30.1	77.6	29.0	74.0	27.4	30.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.1	6.8	5.5	DB	-16.0	34.5	2.2	1.5	-17.6	35.5	-18.9	36.4	-20.1	37.2	-21.7	38.3	(4)
(5)			WB	-16.9	27.6	2.0	1.3	-18.3	28.6	-19.5	29.3	-20.6	30.1	-22.0	31.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.1	-3.4	-0.9	4.4	10.8	16.2	20.1	23.5	23.9	19.3	13.3	5.6	-0.8	(6)
(7)		DBStd	10.01	3.45	3.50	3.70	3.94	3.46	3.00	3.03	2.37	2.55	3.22	4.11	3.69	(7)
(8)		HDD10.0	1419	414	306	179	36	1	0	0	0	0	9	140	334	(8)
(9)		HDD18.3	3123	673	539	433	227	85	16	1	1	18	157	382	592	(9)
(10)		CDD10.0	1808	0	0	5	61	192	303	418	430	278	113	8	0	(10)
(11)		CDD18.3	470	0	0	0	2	18	68	162	172	47	1	0	0	(11)
(12)		CDH23.3	3227	0	0	0	33	223	465	1167	1117	214	7	0	0	(12)
(13)		CDH26.7	941	0	0	0	4	56	139	406	301	34	0	0	0	(13)
(14)	Wind	WSAvg	1.6	1.8	1.8	1.9	1.9	1.8	1.5	1.3	1.2	1.3	1.5	1.6	1.8	(14)
(15)	Precipitation	PrecAvg	852	13	18	20	48	66	96	213	169	111	44	43	19	(15)
(16)		PrecMax	1273	42	86	82	137	140	237	493	544	292	119	237	133	(16)
(17)		PrecMin	417	0	0	4	4	11	18	13	27	17	2	0	0	(17)
(18)		PrecStd	223	10	22	16	34	38	59	114	113	73	31	49	26	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.8	11.7	20.1	26.9	30.9	33.2	34.4	33.3	29.9	24.3	18.9	10.8	(19)
(20)			MCWB	3.1	6.1	12.1	15.2	18.8	21.0	26.1	26.1	22.6	17.0	12.1	5.9	(20)
(21)		2%	DB	5.5	9.2	16.2	23.7	28.1	30.1	32.5	31.4	27.4	22.2	16.1	8.5	(21)
(22)			MCWB	1.5	4.3	9.0	13.6	17.4	20.9	25.1	25.5	20.6	16.0	10.7	4.3	(22)
(23)		5%	DB	3.9	7.0	13.6	21.1	25.8	27.9	30.8	29.6	25.7	20.6	14.1	6.7	(23)
(24)			MCWB	0.6	2.7	7.3	12.6	16.6	20.5	24.6	24.8	20.1	15.0	9.3	3.3	(24)
(25)		10%	DB	2.3	4.9	11.2	18.8	23.6	25.9	29.1	28.2	24.2	19.1	12.1	5.1	(25)
(26)			MCWB	-0.6	1.3	5.8	11.1	15.7	19.8	24.1	23.9	19.5	14.3	8.3	1.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.9	6.8	12.9	16.6	21.2	23.8	28.1	27.8	24.1	19.1	13.2	6.9	(27)
(28)			MCDWB	6.9	10.5	18.5	23.2	27.3	28.8	32.1	31.4	28.4	22.1	16.8	9.2	(28)
(29)		2%	WB	2.2	4.6	9.7	14.7	19.1	22.3	26.3	26.4	22.6	17.3	11.6	5.1	(29)
(30)			MCDWB	4.6	8.3	14.6	21.8	26.1	27.6	30.5	29.8	25.5	20.5	15.0	7.7	(30)
(31)		5%	WB	0.9	3.0	7.8	13.2	17.7	21.3	25.3	25.5	21.5	16.1	10.2	3.5	(31)
(32)			MCDWB	3.2	6.4	12.7	19.8	23.8	25.8	29.3	28.7	24.0	19.2	13.3	6.2	(32)
(33)		10%	WB	-0.1	1.6	6.2	11.8	16.5	20.4	24.5	24.6	20.4	15.1	8.8	2.2	(33)
(34)			MCDWB	1.9	4.3	10.7	17.7	22.0	24.4	27.9	27.3	23.1	18.0	11.7	4.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.9	9.8	9.6	10.8	10.2	8.0	6.5	6.9	8.7	9.9	9.1	9.2	(35)
(36)			MCDBR	11.1	12.2	13.7	15.3	15.6	12.6	10.3	9.2	11.1	12.1	12.0	10.7	(36)
(37)			MCWBR	8.3	8.5	8.2	7.9	7.5	5.7	4.7	4.5	5.6	6.9	7.8	7.8	(37)
(38)		5% WB	MCDWR	10.2	11.1	12.7	13.9	13.1	10.1	8.8	8.0	8.7	10.2	10.5	9.9	(38)
(39)			MCWBR	8.0	8.1	8.4	8.1	7.3	5.2	4.6	4.5	5.2	6.7	7.3	7.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.382	0.450	0.527	0.602	0.626	0.675	0.620	0.547	0.486	0.490	0.438	0.382	(40)	
(41)		taud	2.021	1.883	1.753	1.656	1.663	1.635	1.811	1.987	2.067	1.934	1.987	2.057	(41)	
(42)		Ebn at Noon	735	731	717	696	693	659	693	734	752	684	664	698	(42)	
(43)		Edh at Noon	122	162	206	242	245	253	210	172	149	153	126	110	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.52	3.42	4.40	5.05	5.45	4.98	4.32	4.28	4.07	3.40	2.39	2.17	(44)	
(45)		RadStd	0.17	0.25	0.35	0.42	0.55	0.49	0.58	0.42	0.42	0.28	0.33	0.18	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (18 neighbors)		N/A	N/A	N/A	N/A	+0.37	+0.35	N/A	N/A	+82	+42

Nomenclature: See separate page